

Nirmal Patel

Patan, Gujarat, India | +91-8140637181 | nilupatel02005@gmail.com | LinkedIn | Portfolio | GitHub

PROFILE

AI/ML-focused developer with hands-on experience building real-world projects in **Deep Learning, NLP, and model deployment** using **FastAPI**. Strong foundation in **data preprocessing, feature engineering, model training & evaluation**, and deploying ML systems into usable web apps. Comfortable working across the full pipeline from dataset preparation to inference APIs while writing clean, scalable Python code.

SKILLS

Languages: Python, PHP, SQL

Frameworks & Libraries: TensorFlow, Keras, Pandas, Matplotlib, Scikit-learn, Laravel

Machine Learning & Deep Learning Concepts: Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Model Training & Evaluation, Hyperparameter Tuning, Supervised & Unsupervised Learning, Classification, Regression, ANN, CNN, RNN, LSTM, NLP, Transfer Learning, Transformers

Database & Tools: MySQL, Git, VS Code, Jupyter Notebook, Postman, Hugging Face

EDUCATION

Ganpat University

Gujarat, India

B.Sc. (CA & IT) | 2022 – 2025 | CGPA: 7.97

M.Sc. (CA & IT) | 2025 – 2027 | Current SGPA: 8.6

EXPERIENCE

Backend Developer Intern - BM Coder

Jan 2025 – Jun 2025

- Designed and developed **RESTful APIs** using Laravel following MVC architecture.
- Built optimized **MySQL schemas** and improved query performance for high-usage workflows.
- Implemented **authentication, validation, error handling**, and secure CRUD operations.
- Debugged and resolved production-level issues, improving stability and reducing repeated failures.
- Worked in an Agile environment: **code reviews, sprint tasks, testing, and collaboration** with senior developers.

PROJECTS

CommentVision AI – YouTube Toxicity Analyzer

Sep 2025 | [Live Demo Of Project](#)

- Trained a **BiGRU multi-label model** to detect **6 toxicity categories** using the **Jigsaw Toxic Comment Dataset**.
- Built a **FastAPI web app** with Jinja2 + Chart.js to show toxicity counts, charts, and top flagged comments.
- Optimized prediction flow to process multiple comments together for faster analysis.
- Integrated YouTube comment extraction to scan **up to 500 comments in <8 seconds on CPU**.
- Tech:** Python, TensorFlow/Keras, Deep Learning, NLP, FastAPI, YouTube API, Chart.js

CarDamageAnalysis AI – Multi-Stage Car Damage Assessment System

Aug 2025 | [Live Demo Of Project](#)

- Built a **3-stage EfficientNet-B3 pipeline** for car detection, damage type classification, and severity grading.
- Trained on **67K+ real-world images** with strong augmentation for better performance on different angles/lighting.
- Achieved strong accuracy across stages: **99.1% / 97.8% / 94.6%**.
- Deployed a live app that predicts damage output with confidence scores from a single uploaded image.
- Tech:** Python, TensorFlow/Keras, CNN, EfficientNet, Transfer Learning

Used Car Value Estimator — Resale Price Prediction

Jul 2025 | [Live Demo Of Project](#)

- Developed a Neural Network model to estimate car resale value using attributes like **age, mileage, and brand**.
- Performed full preprocessing: **cleaning, encoding, scaling**, and feature handling for stable predictions.
- Evaluated model performance and improved results through **training tuning + normalization**.
- Deployed the model via **FastAPI** and hosted the app on **Hugging Face** for real-time predictions.
- Tech:** Python, Feature Engineering, TensorFlow, ANN, Pandas, NumPy, FastAPI, Hugging Face